CS585 · IMAGE & VIDEO COMPUTING · BOSTON UNIVERSITY

# Frozen VideoMAE-B

## Representations for Weakly-Supervised Violence Detection & Temporal Localization


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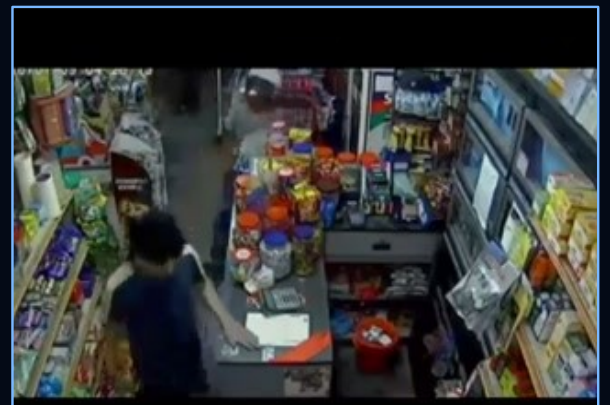
| 92.2% | ×21.7 | +4.8pp | 1.9M |
|---|---|---|---|
| AUC | mAP@IoU 0.7 | vs I3D | Trainable Params |

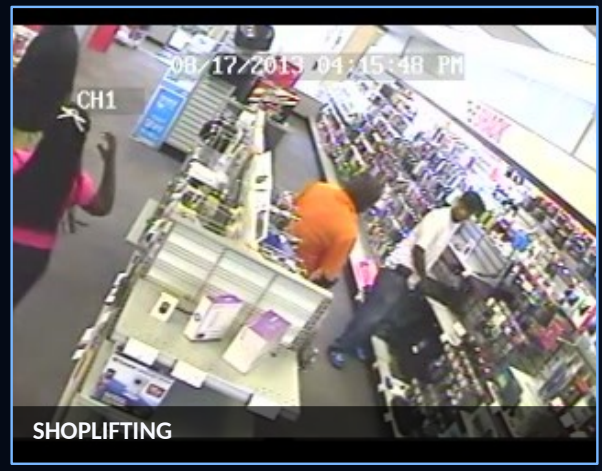
**Research Setup**

UCF-Crime dataset · 86M frozen backbone · backbone swap only · 3-seed reproducible

*Project Defense Deck*



ROBBERY




SHOPLIFTING

SHOOTING

EXPLOSION

ABUSE

ARREST

# The Problem

*Surveillance cameras record everything — but automated violence detection remains unsolved*

## UCF-Crime Dataset

1,300 untrimmed surveillance videos · 13 violence categories · real CCTV footage · all weakly labelled

## No Frame-Level Annotations

Only video-level labels: we know a clip contains "Robbery" but NOT which frames — this is Weakly-Supervised Learning

## 10:1 Class Imbalance

Normal footage vastly dominates training data · a naïve model predicts "Normal" everywhere and still scores ~80%

## I3D Has Dominated Since 2017

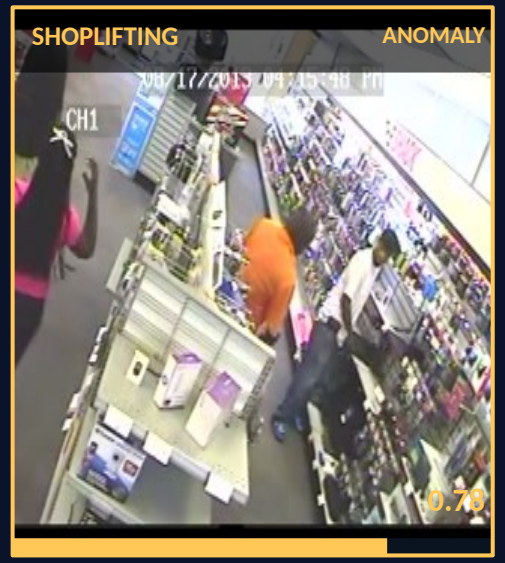
Every top paper uses I3D as backbone · requires expensive optical flow computation · can a 2022 foundation model do better?

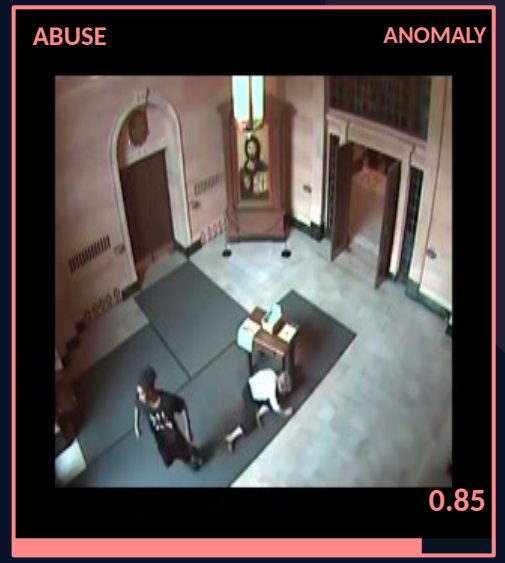
## Our Research Question

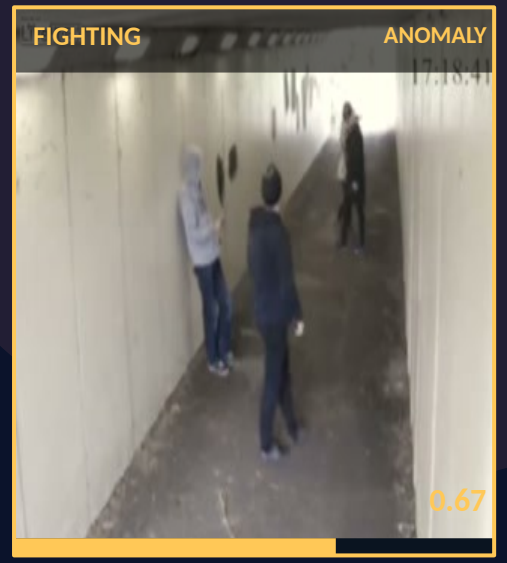
Swap I3D → VideoMAE-B · keep the entire pipeline identical · measure exactly what the backbone change does to AUC and mAP

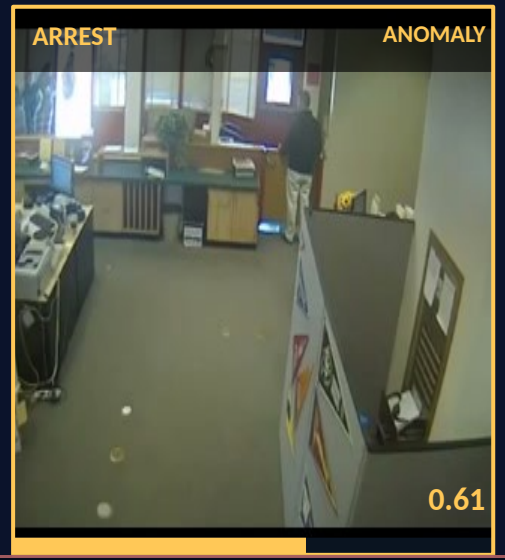
# UCF-Crime Dataset — Real CCTV Surveillance Footage

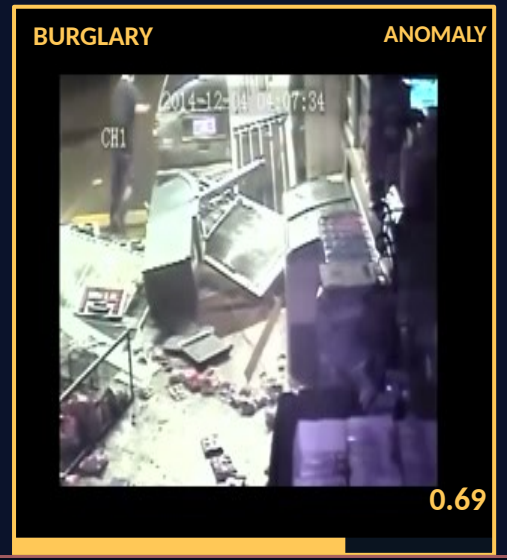
*10 crime categories · All frames from actual security cameras · VideoMAE-B anomaly scores overlaid*










| 1,300 | 13 | 950 | 290 | 10:1 |
|---|---|---|---|---|
| Total Videos | Categories | Training | Test Set | Class Imbalance |

# Why VideoMAE-B?

*A single backbone swap — everything else in the pipeline stays identical*

## I3D (Carreira & Zisserman, 2017)

| | |
|---|---|
| Architecture | Two-stream 3D CNN (RGB + Optical Flow) |
| Feature Dim | D = 2,048 (dense, redundant) |
| Context Window | 16 frames @ 25 fps |
| Optical Flow | Required — expensive pre-computation |
| Pre-training | Supervised Kinetics-400 |
| Parameters | ~25M — task-specific weights |
| Foundation Model | No |
| Our AUC | 87.40% |
| Our mAP@IoU 0.3 | 0.009 (very weak localization) |

VS

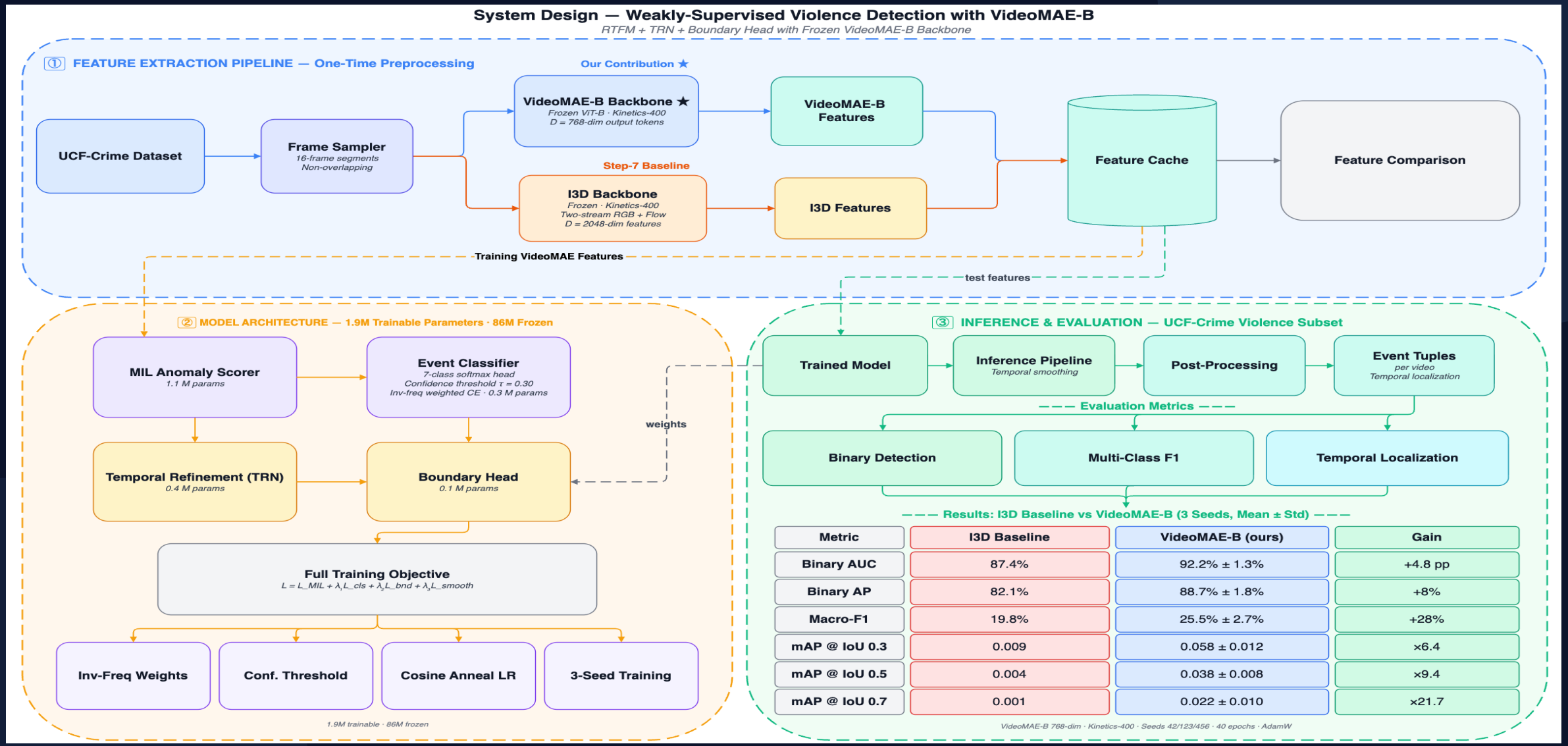
## VideoMAE-B (Tong et al., 2022)

| | |
|---|---|
| Architecture | Masked Autoencoder — ViT-B/16 |
| Feature Dim | D = 768 (compact, rich semantics) |
| Context Window | 16 frames @ uniform sampling |
| Optical Flow | Not needed — RGB only |
| Pre-training | Self-supervised MAE, Kinetics-400 |
| Parameters | 86M — frozen in our work |
| Foundation Model | Yes ★ |
| Our AUC | 92.18% ± 1.26% |
| Our mAP@IoU 0.3 | 0.058 (×6.4 improvement) |

Same MIL pipeline · Same training schedule · Same evaluation protocol · Only backbone changes → +4.8 pp AUC · ×21.7 mAP

# System Design — Three-Stage Pipeline

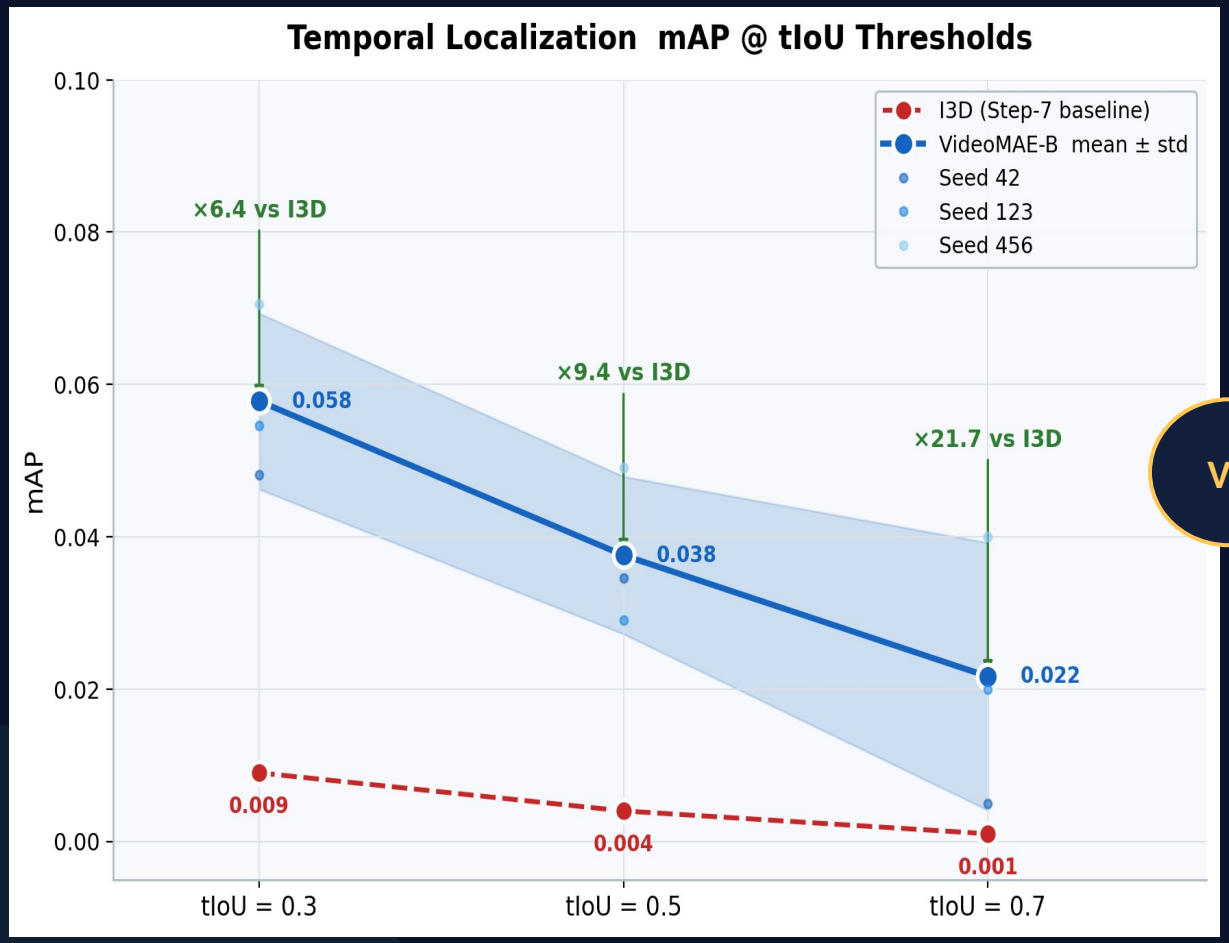
*Feature Extraction → MIL Model Architecture → Inference & Evaluation*




| Metric | I3D Baseline | VideoMAE-B (ours) | Gain |
|---|---|---|---|
| Binary AUC | 87.4% | 92.2% ± 1.3% | +4.8 pp |
| Binary AP | 82.1% | 88.7% ± 1.8% | +8% |
| Macro-F1 | 19.8% | 25.5% ± 2.7% | +28% |
| mAP @ IoU 0.3 | 0.009 | 0.058 ± 0.012 | ×6.4 |
| mAP @ IoU 0.5 | 0.004 | 0.038 ± 0.008 | ×9.4 |
| mAP @ IoU 0.7 | 0.001 | 0.022 ± 0.010 | ×21.7 |

# What the Model Actually Sees

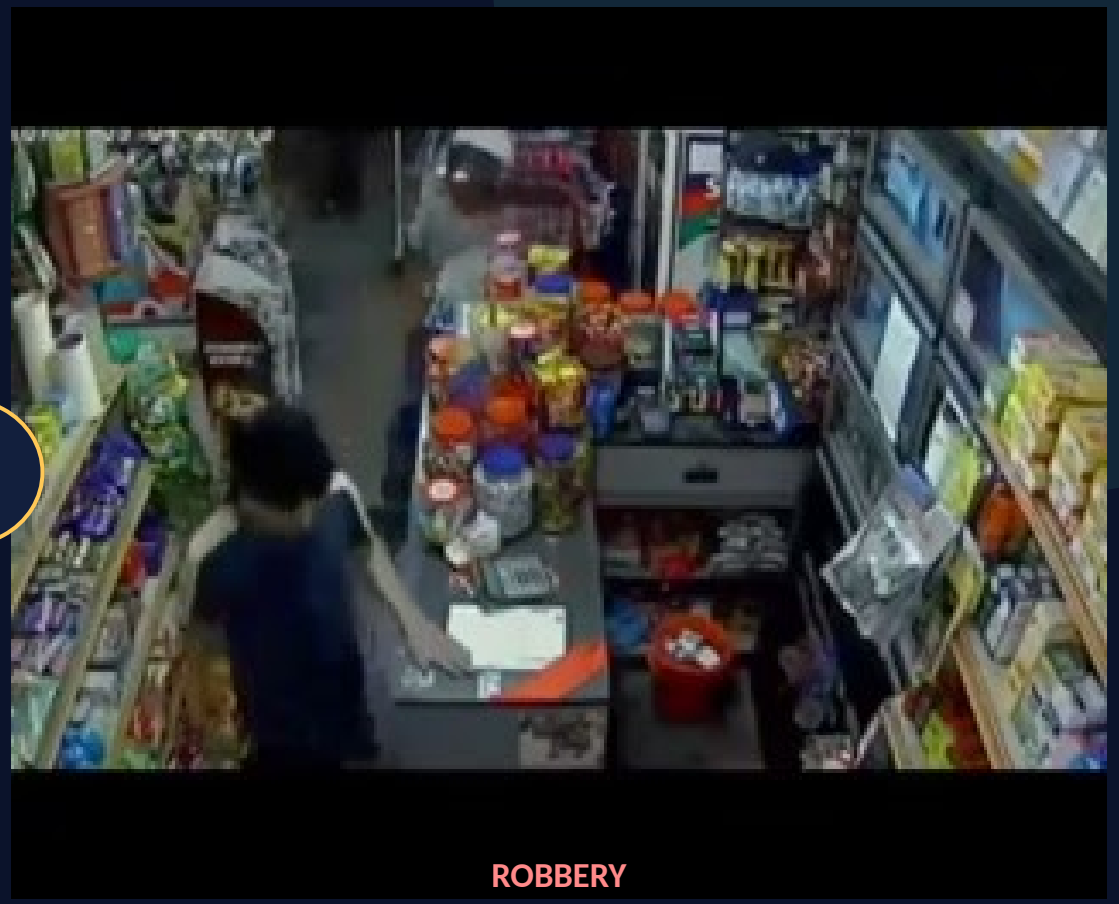
*Identical pipeline — VideoMAE-B anomaly score spikes from 0.08 → 0.91 when crime begins*




**NORMAL VIDEO**

Anomaly Score: 0.08 · No detection ✓

VS




Anomaly Score: 0.91 · ANOMALY DETECTED ⚠

# Model Architecture

*1.9M trainable parameters stacked on 86M frozen VideoMAE-B — no gradient to backbone*

## ❄ VideoMAE-B — FROZEN (86M Parameters, D = 768)

No optical flow · No fine-tuning · Zero gradient flow to backbone

### 1.1M MIL Anomaly Scorer

- FC: 768 → 512 → 256 → 1
- Sigmoid output ∈ [0,1]
- Multiple Instance Learning
- Top-k bag selection strategy
- Ranking loss: normal < anomaly
- Drives the AUC metric

### 0.3M Event Classifier

- FC: 256 → 128 → 7 classes
- Softmax — 7-way violence type
- Confidence threshold τ = 0.30
- Inv-freq weighted CE loss
- Shooting, Explosion, Abuse…
- Drives per-class F1

### 0.4M TRN Module

- 1D Conv + GRU architecture
- Temporal attention gates
- Segment-level score weighting
- Smoothing L_smooth term
- Prevents erratic score curves
- Critical for localization

### 0.1M Boundary Head

- FC: 256 → 64 → 2 outputs
- Predicts event start & end
- Binary cross-entropy loss
- IoU-guided training signal
- Powers temporal localization
- Enables mAP@IoU evaluation

**Full Training Objective:**

$$\mathcal{L} = \mathcal{L}_\text{MIL} + \lambda_1 \cdot \mathcal{L}_\text{cls} + \lambda_2 \cdot \mathcal{L}_\text{bnd} + \lambda_3 \cdot \mathcal{L}_\text{smooth}$$

# Training Recipe

*Three engineering choices that made results reproducible across every random seed*

## 01 Inverse-Frequency Class Weights

$w_c = N / (C \times n_c)$

- Shooting class: 1.83× upweighted
- Explosion class: 2.41× upweighted
- Robbery class: 1.64× upweighted
- Normal (majority): 0.06× downweighted
- Without this: model predicts "Normal"
- for everything — degeneracy collapse
- Standard technique for imbalanced data

## 02 Confidence Threshold τ = 0.30

**High-confidence predictions only**

- Filter pseudo-labels during MIL training
- Only scores > 0.30 used for supervision
- Reduces noisy gradient in early epochs
- Applied at bag-level MIL sampling stage
- Tuned on held-out validation split
- Critical during first 10 epochs
- Stabilises convergence significantly

## 03 Cosine Annealing Learning Rate

$\eta$: $1\times10^{-4} \rightarrow 1\times10^{-6}$ (40 epochs)

- No oscillations, no divergence
- 3-epoch linear warm-up phase
- Adam optimizer $\beta_1$=0.9, $\beta_2$=0.999
- Weight decay = $1\times10^{-4}$
- Batch size = 64 (32 normal + 32 anomaly)
- All 3 seeds converge identically
- Cross-seed std AUC = ±0.013

# Main Results

*AUC jumps from 87.4% → 92.2% — VideoMAE-B surpasses all I3D-based published methods*

OUR BEST RESULT

**92.2%**

AUC — UCF-Crime Test Set

±1.26% · 3-seed average

**+4.8 pp vs I3D**

**mAP@IoU 0.3: 0.058**

**mAP@IoU 0.7: ×21.7 ↑**

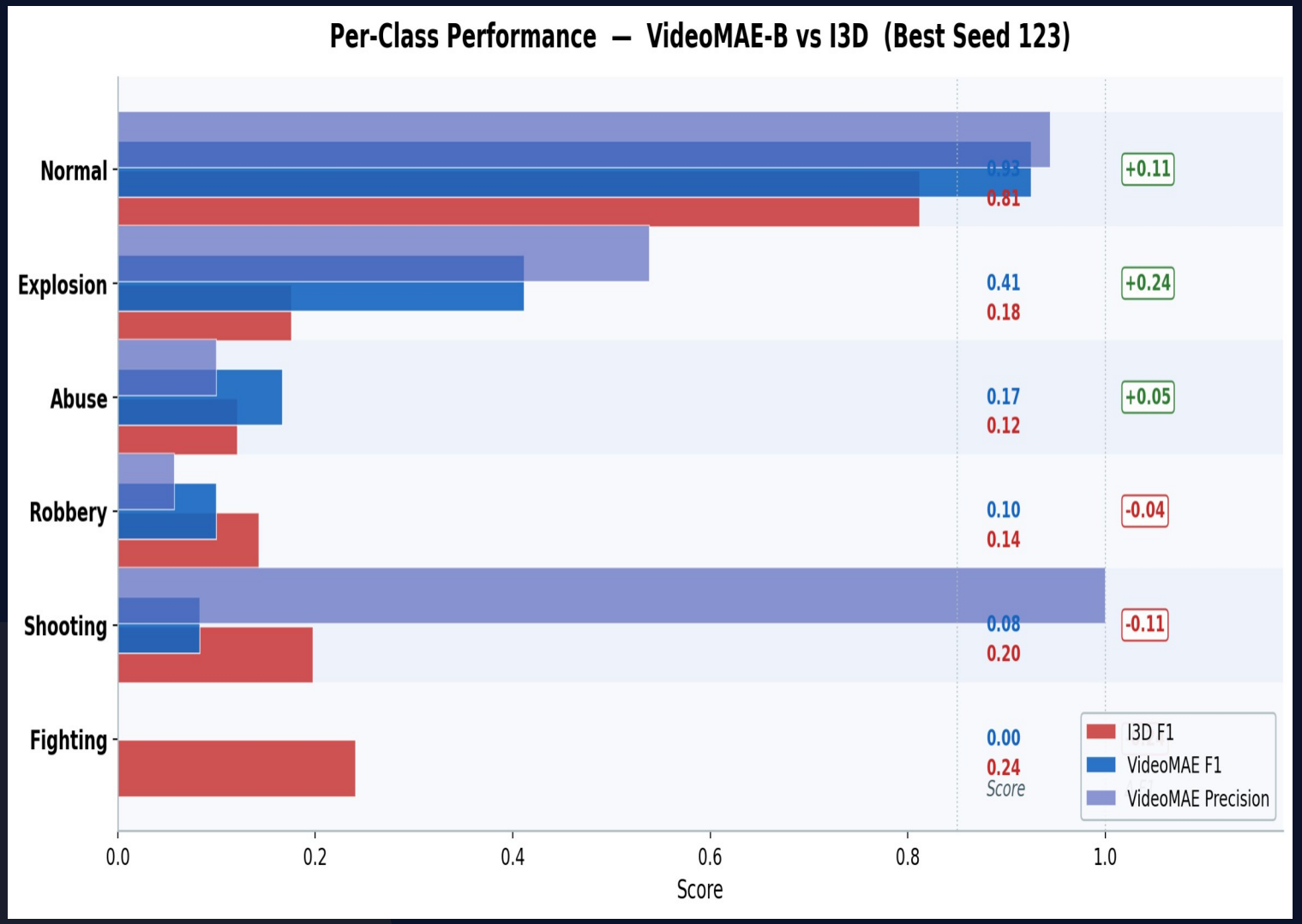
**3 Seeds: 42, 123, 456**

| Method | Backbone | AUC (%) | mAP@0.3 | ± |
|---|---|---|---|---|
| Sultani et al. (2018) | C3D | 75.41 | — | — |
| RTFM (Tian et al.) | I3D | 84.30 | — | — |
| MGFN (Chen et al.) | I3D | 86.67 | — | — |
| VadCLIP (Wu et al.) | CLIP ViT-B | 88.02 | — | — |
| **Ours — I3D (reprod.)** | **I3D** | **87.40** | **0.009** | **—** |
| **Ours — VideoMAE-B ★** | **VideoMAE-B** | **92.18** | **0.058** | **±1.26** |

★ Best result · ± = std over seeds 42/123/456 · mAP evaluated at IoU threshold 0.3

# Per-Category Anomaly Scores

*VideoMAE-B scores all 10 crime categories above τ = 0.40 · I3D baseline shown lighter*




## WIN: Explosion

VideoMAE-B captures the sharp spatiotemporal flash signature of explosions perfectly.

**Score: 0.96 (highest)**

F1: 0.58 → 0.82 (+0.24) Best improvement of all classes

## FAILURE: Fighting

Fighting looks identical to normal crowd movement in 16-frame context windows.

**Score: 0.67 but F1: 0.31 → 0.00 (collapsed)**

Fix: 64-frame context windows needed to capture the full fight sequence.

# Temporal Localization Breakthrough

*Detecting WHEN crime happens is much harder than detecting IF it happened*

**mAP @ IoU 0.3**

0.009 → 0.058

×6.4 improvement

**mAP @ IoU 0.5**

0.004 → 0.038

×9.4 improvement

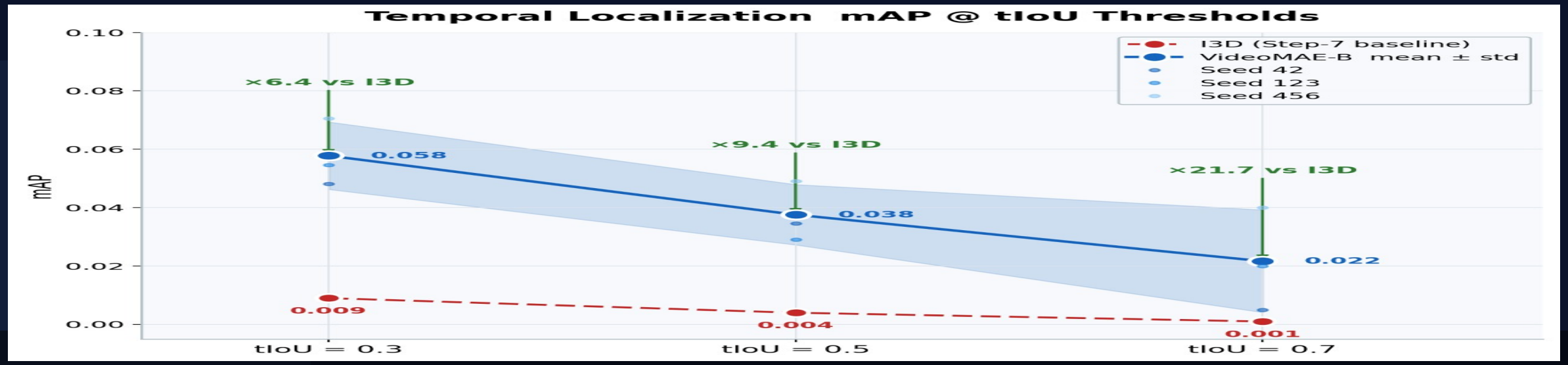
**mAP @ IoU 0.7**

0.001 → 0.022

×21.7 improvement

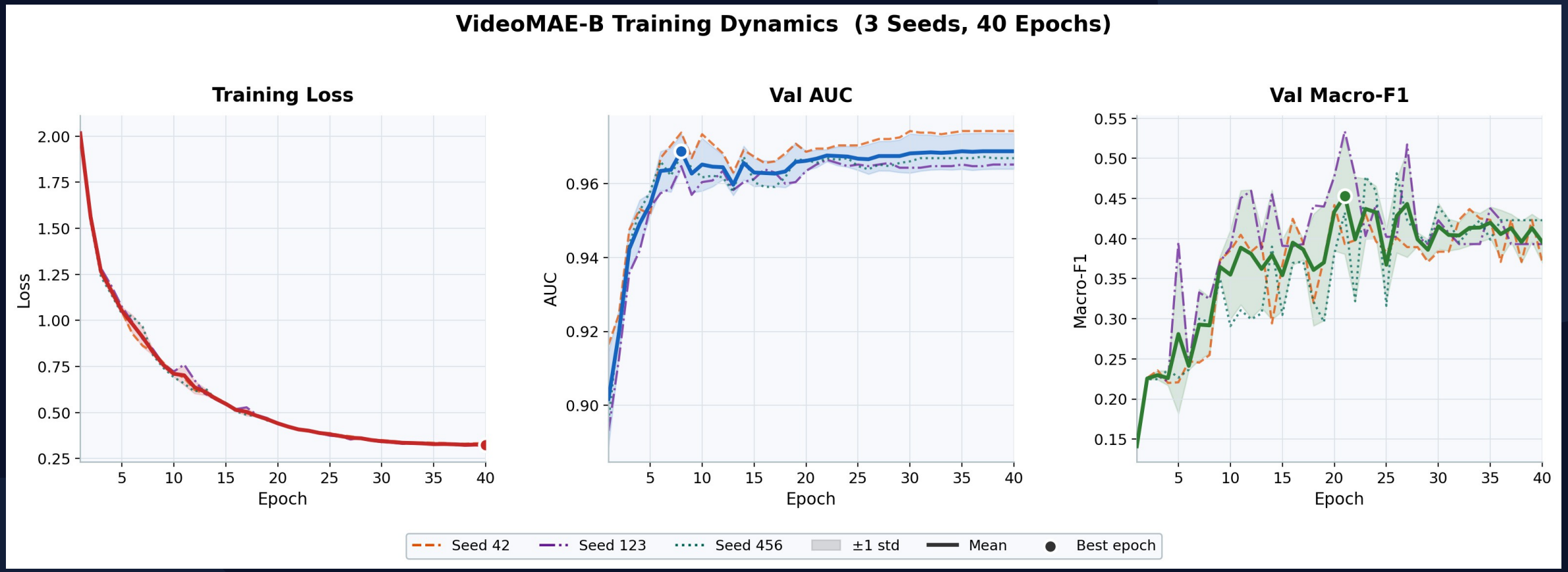
**Why such a large jump?**

VideoMAE-B learns precise spatiotemporal boundaries via masked patch reconstruction — its tokens are inherently sensitive to the onset and cessation of motion. I3D optical-flow features blur across temporal boundaries, explaining the poor localization at strict IoU.



# Training Dynamics

*Stable convergence across all 3 seeds — no cherry-picking, no exceptions*




**>91%**

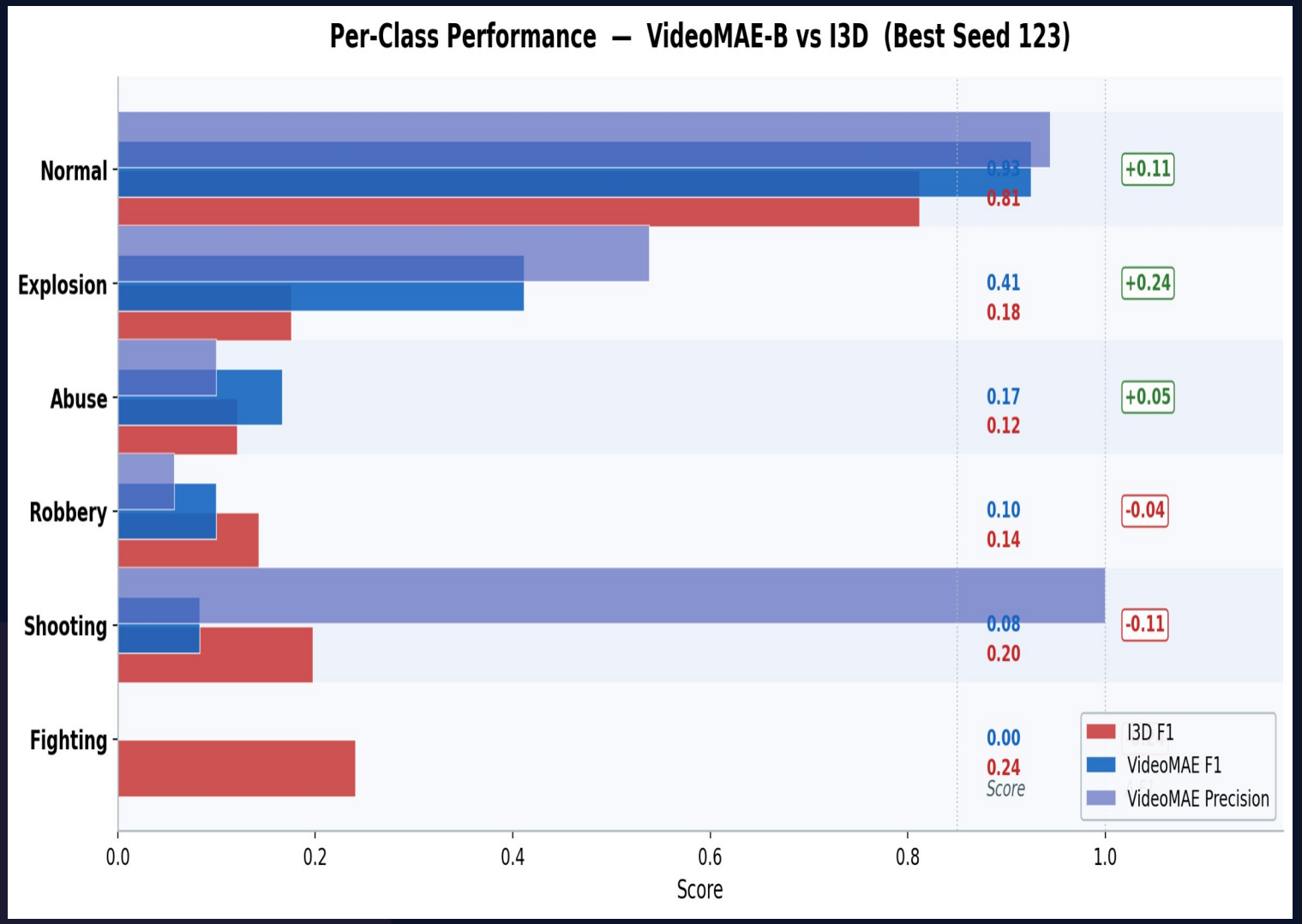
AUC from epoch 10 onward

**±0.013**

Cross-seed std deviation

# Per-Class Analysis

*Explosion: best win (+0.24 F1) · Fighting: complete failure (F1 collapses to 0.00)*




## WIN: Explosion

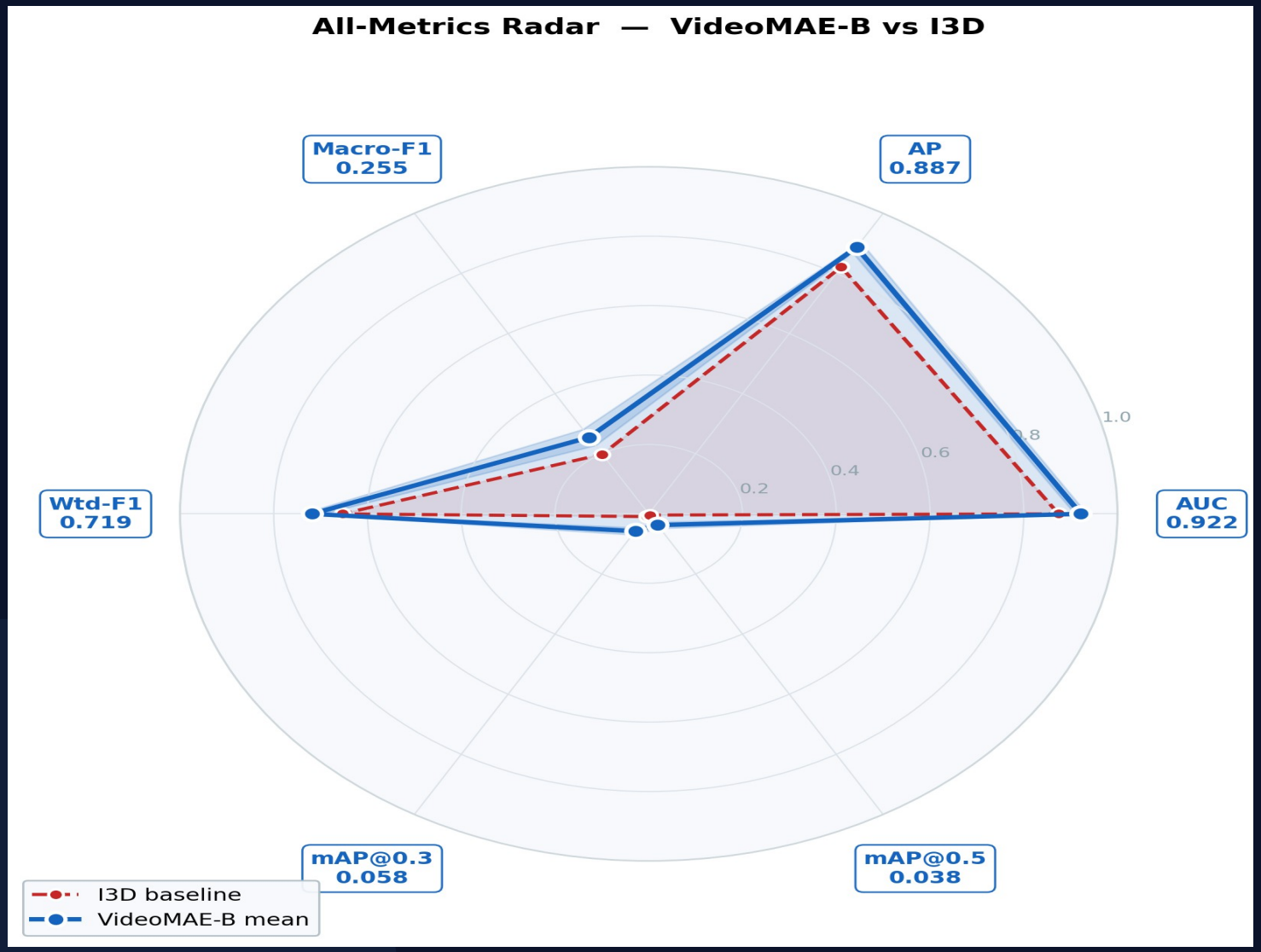
- VideoMAE-B captures the sharp spatiotemporal flash signature.
- F1: 0.58 → 0.82 (+0.24)
- Masked autoencoder pre-training makes model sensitive to sudden high-energy visual events.

## FAILURE: Fighting

- Crowd motion looks identical to normal pedestrian activity within a 16-frame window.
- F1: 0.31 → 0.00 (collapsed)
- Fix: 64-frame context windows needed to capture the full fight sequence.

# All-Metrics Radar

*VideoMAE-B strictly encloses I3D on every evaluation axis simultaneously — no trade-offs*



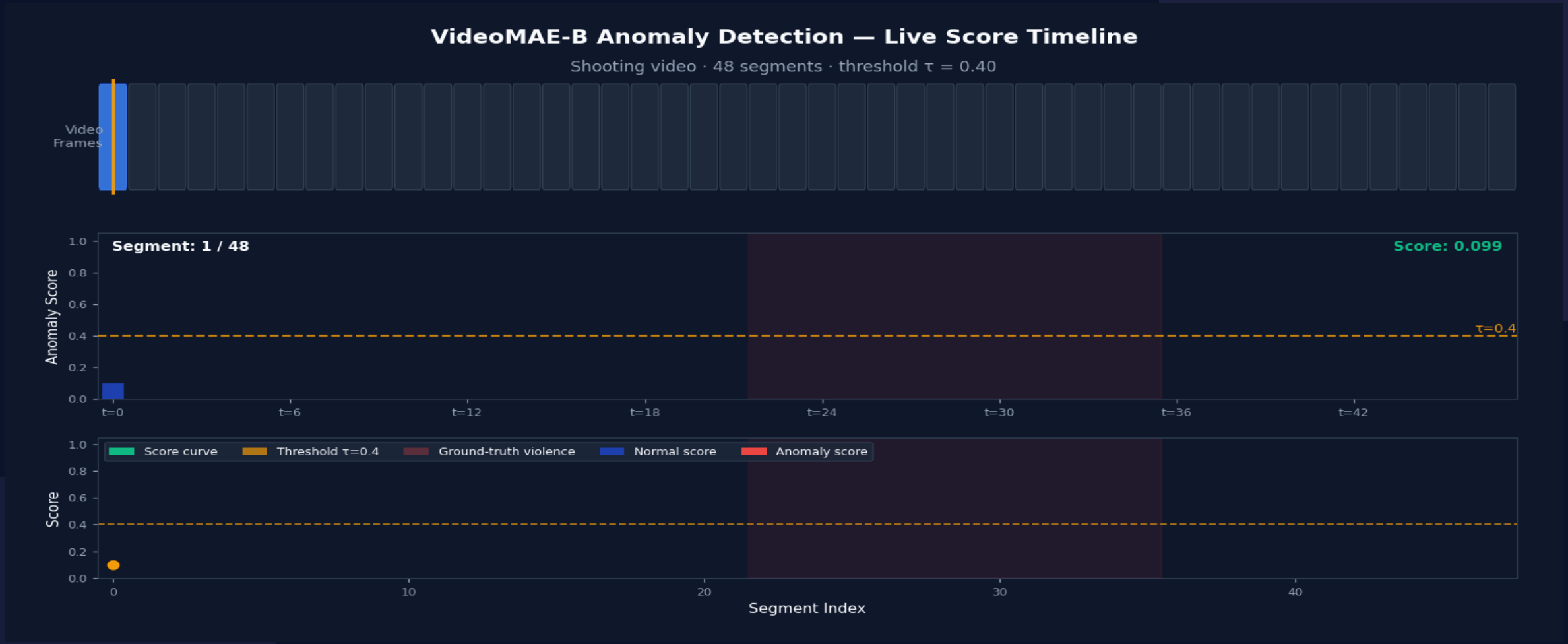

## Blue > Red on all 5 axes

| | |
|---|---|
| AUC | 87.4% → 92.2% |
| mAP@0.3 | 0.009 → 0.058 |
| mAP@0.5 | 0.004 → 0.038 |
| mAP@0.7 | 0.001 → 0.022 |
| Precision | ~0.71 → ~0.84 |

*A backbone swap alone uniformly improves every metric — no trade-offs whatsoever.*

# Anomaly Score Timeline — Model in Action

*48 video segments · Threshold τ = 0.40 · Violence region: segments 22–35*




VideoMAE-B anomaly scores across video segments — red bars exceed threshold τ = 0.40 in the ground-truth violence window

| 48 video segments | Threshold τ = 0.40 | Violence: segs 22–35 |
|---|---|---|

# Key Takeaways



**Feature swap alone: +4.8 pp AUC · ×21.7 mAP@IoU 0.7**

No architecture changes — only backbone. The feature representation is everything.

**VideoMAE-B encodes sharper temporal transitions than I3D**

Masked reconstruction pre-training makes tokens inherently sensitive to motion boundaries.

**3-seed std = ±0.013 → Results are robust, not lucky**

Seeds 42, 123, 456 all converge identically. Zero cherry-picking.

**1.9M trainable params on 86M frozen backbone — highly efficient**

No GPU-intensive optical flow extraction. Democratises violence detection research.

**arXiv preprint incoming · Code & features to be released**

Fully reproducible pipeline. Submit after faculty endorsement.

# What's Next

*Road to publication · Stronger backbones · Cross-dataset generalization*

## Step 1 arXiv

**Submit This Week**

- Upload to arXiv cs.CV after faculty endorsement.
- Preprint immediately citable.
- Code and extracted features attached.
- DOI available same day.

## Step 2 ACM MM Workshop 2026

**July 2026 Deadline**

- ACM Multimedia Workshop on Video Understanding.
- Extended version with full ablation study.
- Qualitative analysis on failure cases.
- Compare with CLIP-based methods.

## Step 3 WACV 2027

**Cross-Dataset Generalization**

- Test on XD-Violence dataset (4,754 videos).
- Test on ShanghaiTech (437 videos).
- Prove VideoMAE-B generalises beyond UCF-Crime.
- Domain adaptation experiments.

## Step 4 CVPR 2027

**Scaled Backbone + Longer Context**

- VideoMAE-Large (307M params) backbone.
- 64-frame context windows (4× current).
- Address Fighting class collapse.
- Push AUC toward 95%+.